C3.1 Integration of body systems: data-based questions

NAME	DATE	TOTAL
		_____ / 8

Read the data carefully before you answer. The number of marks shows how many separate points to make. Show your working in calculations.

1 Body clocks in mice

8 MARKS

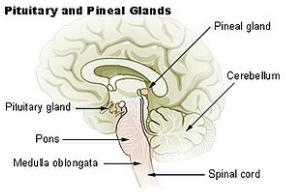
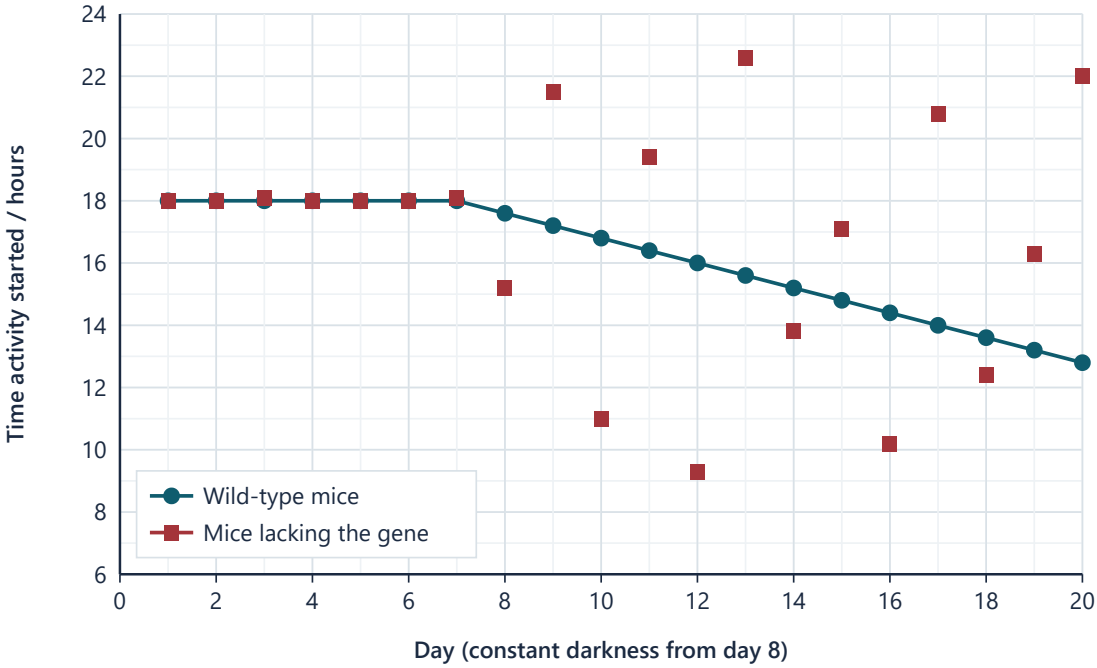


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Mice were kept with 12 hours of light and 12 hours of dark (lights off at 18:00) for 7 days, then in constant darkness for 13 days. The time each mouse started running on its wheel was recorded every day. Normal (wild-type) mice were compared with mice lacking a gene for a clock protein.



a STATE when both groups of mice became active during the first 7 days. [1]

b SUGGEST one advantage to mice of being active at night. [1]

c CALCULATE the length of one daily cycle (the period) of the wild-type mice in constant darkness. Show your working. [2]

d **COMPARE** the two groups in constant darkness. [2]

e **EVALUATE** the hypothesis that this gene is needed for the body clock. [2]